Scenario 2 — Heterogeneous Backend Latency Analysis

1 Scenario Objective

This scenario tests whether routing strategies respond appropriately when backend service times differ substantially. The three backends are configured with fixed latencies of 50 ms, 100 ms, and 200 ms, while the load, concurrency, and measurement window remain comparable to the baseline experiment.

The equal-latency baseline could not reveal whether a strategy reacts to backend performance, because every backend was equally attractive. In this scenario, a latency-blind policy should continue to distribute work according to its static rule, whereas a feedback-based policy can reduce average latency by directing more traffic toward the faster backends.

The configured weights remain 3:1:2, so they do not encode the latency ordering: Backend 1 is both the fastest and highest-weighted, while Backend 3 is the slowest and has weight 2. Weighted Round Robin therefore still sends one third of requests to the slowest backend. This makes the scenario useful for separating the effect of static weights from the effect of latency-aware routing.

2 Experimental Configuration

Table 1: Heterogeneous-latency configuration.

Parameter	Value
Backends	3
Backend latencies	50 ms, 100 ms, 200 ms
Configured weights	3 : 1 : 2
Request rate	150 requests/s (open loop)
Concurrency	32 workers
Warmup	5 s
Measurement duration	30 s
Repetitions	5 per strategy
Requests per run	4,500
Total runs	20

Every run completed with exactly 4,500 measured requests, giving 22,500 requests per strategy. The summary contains 20 results and every summary entry matches its corresponding individual JSON file exactly.

3 Results by Strategy

Definitions: **median** and **IQR** are computed over the five per-run mean latencies. **P50/P95/P99** are medians of the per-run percentile values. **Pooled mean** is the latency mean over all requests for that strategy.

Table 2: Latency summary by strategy. Median and IQR are computed over the five per-run mean latencies; P50/P95/P99 are medians of the per-run percentile values.

Strategy	Median (ms)	IQR (ms)	Pooled mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)	Errors
Round Robin	**118.681**	0.216	118.616	102.1	202.3	202.7	0 / 22,500
Least Connections	**84.603**	0.458	84.631	52.6	202.2	202.7	0 / 22,500
Weighted Round Robin	**110.386**	0.057	110.414	54.3	202.5	202.9	0 / 22,500
Adaptive	**84.879**	0.140	84.847	52.6	202.2	202.7	0 / 22,500

Table 3: Per-run mean latency by strategy (ms).

Strategy	Run 1	Run 2	Run 3	Run 4	Run 5
Round Robin	118.493	118.709	118.681	118.464	118.735
Least Connections	84.603	84.897	85.589	84.439	83.627
Weighted Round Robin	110.376	110.440	110.386	110.486	110.383
Adaptive	85.034	84.879	84.941	84.581	84.800

Round Robin distributes requests exactly evenly and Weighted Round Robin exactly follows the configured 3:1:2 weights. Both feedback-based strategies concentrate traffic on the two faster backends: Least Connections and Adaptive send only about 12.5% of requests to the 200 ms backend, compared with 33.3% for the static strategies.

4 Comparative Analysis

Least Connections and Adaptive clearly outperform the static strategies on mean latency. Using the median run means, Adaptive is about 28.5% faster than Round Robin and about 23.1% faster than Weighted Round Robin. Least Connections achieves a similar reduction: about 28.7% versus Round Robin and about 23.4% versus Weighted Round Robin. These effects are tens of milliseconds and are far larger than the run-to-run variation observed within any strategy.

Least Connections records the lowest point estimate (median 84.603 ms), while Adaptive records 84.879 ms, a difference of about 0.28 ms or 0.3%. Their pooled means differ by about 0.22 ms. This difference is of the same order as run-to-run variability and is not practically meaningful. In this experiment, Adaptive matches Least Connections but does not demonstrate a clear advantage over it.

Weighted Round Robin also improves on Round Robin, reducing mean latency by about 7.0%. This follows from its distribution: it sends 50% of requests to the 50 ms backend, whereas Round Robin sends only one third there. It remains substantially slower than the feedback-based strategies because it still sends one third of requests to the 200 ms backend.

The percentile results tell a more nuanced story:

- Round Robin has a P50 of 102.1 ms because its median request lands on the 100 ms backend.
- Weighted Round Robin has a P50 of 54.3 ms because exactly half of its requests go to the 50 ms backend.
- Least Connections and Adaptive have P50 values of about 52.6 ms because roughly 60% of their requests go to the fastest backend.
- All strategies have P95 and P99 values near 202–203 ms. This is expected because every strategy still sends more than 5% of requests to the 200 ms backend, so both the 95th and 99th percentiles are dominated by slow-backend responses.

Thus, latency-aware routing substantially improves the average case, but it does not eliminate exposure to the slow backend's tail latency in this configuration. Adaptive provides no measured P95 or P99 advantage over Least Connections.

5 Benchmark Validity

The experiment is clean and consistent:

- All 20 runs completed with **zero errors** and exactly 4,500 requests per run.
- The summary file matches all individual result files exactly.
- Run-to-run coefficients of variation for mean latency range from approximately 0.04% to 0.84%. Least Connections is the most variable because its decisions depend on dynamic connection state, but its absolute run-mean range is still only about 2 ms.
- Across all runs, P95 remains between 202.0 and 202.5 ms and P99 between 202.6 and 203.0 ms.
- The run-level latency ordering is stable: Least Connections and Adaptive are consistently around 84–85 ms, Weighted Round Robin around 110 ms, and Round Robin around 118–119 ms.

At the window level, the first and last measurement windows show expected completion boundary effects: the first windows contain fewer completed requests and the final windows contain additional requests that were in flight at the deadline. These effects are small and do not affect the aggregate conclusions.

Using the measured backend distributions, the approximate service-time-only means are 116.7 ms for Round Robin, 82.6 ms for Least Connections, 108.3 ms for Weighted Round Robin, and 82.8 ms for Adaptive. The measured means are roughly 2 ms higher in every case, which is consistent with a common harness/proxy overhead. The overhead is similar across strategies and does not affect their comparison.

The effect sizes in this scenario are much larger than the noise observed in the homogeneous baseline, where strategy medians differed by only about 0.06 ms. The heterogeneous experiment is therefore clean enough to support the conclusion that feedback-based routing materially improves mean latency under backend heterogeneity.

6 Conclusion

The heterogeneous-latency scenario shows a clear separation between static and feedback-based routing. With backends fixed at 50, 100, and 200 ms, Round Robin averaged about 118.7 ms and Weighted Round Robin about 110.4 ms, while Least Connections and Adaptive both averaged about 84.6–84.9 ms. Adaptive reduced mean latency by roughly 28.5% relative to Round Robin and 23.1% relative to Weighted Round Robin, but it did not outperform Least Connections, whose median mean was lower by approximately 0.3%. All strategies remained error-free, and P95/P99 stayed near 202–203 ms because every strategy still routed a meaningful share of requests to the 200 ms backend. The measured result therefore supports latency-aware routing as a substantial average-latency improvement over static policies, but does not establish Adaptive as superior to Least Connections in this scenario. The experiment is stable and clean enough to serve as the reference for later degradation, CPU-pressure, and recovery scenarios.